

1.INTRODUCTION:

1.1 Project Overview:

Electric Vehicle Charge and Range Analysis Dashboard

The rapid adoption of Electric Vehicles (EVs) is transforming the global transportation industry by reducing carbon emissions and dependence on fossil fuels. As the number of EV models and charging stations increases, users require an efficient way to compare vehicle performance, charging infrastructure, battery specifications, and market trends. However, EV data is often distributed across multiple datasets, making analysis difficult for customers, manufacturers, and researchers.

The Electric Vehicle Charge and Range Analysis Dashboard is an interactive data visualization project developed using Tableau Desktop. The dashboard integrates four EV-related datasets, including EV population, vehicle specifications, charging stations, and global EV sales. Through charts, maps, KPIs, filters, and story points, users can analyze electric vehicles based on battery capacity, electric range, charging time, manufacturer, and market performance.

The dashboard transforms raw data into meaningful insights, enabling users to make informed decisions regarding electric vehicle selection and market analysis.

1.2 Purpose:

The main purpose of this project is to develop an interactive dashboard that enables users to explore and analyze electric vehicle data efficiently. The dashboard aims to simplify complex datasets using visual analytics, making it easier to compare different EV models, manufacturers, battery capacities, charging infrastructure, and market growth.

The project also demonstrates the use of Business Intelligence tools like Tableau for transforming raw datasets into actionable insights that support decision-making.

Objectives

- Analyze electric vehicle datasets.
- Compare EV brands and models.
- Visualize charging infrastructure.
- Study battery capacity and driving range.
- Analyze global EV sales trends.

- Develop an interactive Tableau dashboard.
- Improve decision-making using data visualization.

2. IDEATION PHASE:

2.1 Problem Statement:

The increasing number of electric vehicles has generated vast amounts of data related to battery specifications, charging stations, manufacturers, and sales. Since this information is scattered across multiple sources, users face difficulty in comparing electric vehicle performance and selecting suitable models.

Existing systems mainly provide static reports or isolated information, making comprehensive analysis difficult. Therefore, there is a need for an interactive visualization platform that combines multiple datasets and presents meaningful insights through dashboards.

2.2 Empathy Map Canvas:

Who are we empathizing with?

- EV Buyers
- Researchers
- Automobile Companies
- Students
- Government Agencies

What do they need?

- Reliable EV information
- Charging station details
- Vehicle comparison
- Market trends
- Interactive visualizations

What do they see?

- Different EV brands
- Various charging technologies

- Multiple battery capacities
- Numerous online reviews

What do they say?

- Which EV has the best driving range?
- Which vehicle charges faster?
- Which brand offers better value?

What do they do?

- Compare specifications
- Visit manufacturer websites
- Search charging station information
- Analyze prices and performance

Pain Points

- Data is scattered.
- Difficult comparison.
- Limited visualization.
- Time-consuming analysis.

Gains

- Easy comparison.
- Better understanding.
- Faster decision-making.
- Interactive dashboard.

2.3 Brainstorming:

Several ideas were considered during project planning:

- EV charging station locator
- EV sales prediction
- Battery health monitoring
- EV recommendation system

- EV visualization dashboard

After evaluating feasibility and available datasets, the team selected the Electric Vehicle Charge and Range Analysis Dashboard because it effectively combines multiple datasets into an interactive analytical platform.

3. REQUIREMENT ANALYSIS:

Requirement Analysis is the process of identifying, analysing, and documenting the functional and technical requirements needed to develop the Electric Vehicle Charge and Range Analysis Dashboard. It helps ensure that the project meets user needs and provides an effective solution for analysing electric vehicle data.

3.1 Customer Journey Map:

A Customer Journey Map illustrates the steps users follow while interacting with the dashboard. It helps understand user behaviour, expectations, challenges, and opportunities for improving the overall experience.

Customer Journey

Stage	User Activity	User Goal	Pain Points	Solution
Awareness	User wants to compare electric vehicles before purchasing.	Find reliable EV information.	Data is scattered across different websites.	Integrate all EV information into one dashboard.
Research	User searches for vehicle specifications and charging station information.	Compare different EV models.	Manual comparison is time-consuming.	Interactive filters and visual comparisons.
Analysis	User analyses battery capacity, driving range, charging time, and sales data.	Identify the most suitable EV.	Large datasets are difficult to interpret.	Charts, KPIs, maps, and dashboards simplify analysis.
Decision	User compares shortlisted vehicles.	Select the best EV.	Limited visualization in existing systems.	Dashboard provides clear comparisons and insights.

Stage	User Activity	User Goal	Pain Points	Solution
Outcome	User makes an informed decision.	Purchase or recommend an EV confidently.	Lack of centralized information.	Complete analysis from a single dashboard.

3.2 Solution Requirements:

Solution requirements describe the essential functionalities and quality attributes that the dashboard must satisfy.

Functional Requirements

1. Import multiple EV datasets into Tableau.
2. Clean and preprocess the data before visualization.
3. Create relationships between datasets.
4. Develop interactive dashboards.
5. Implement filters for brand, model, charging time, battery capacity, and electric range.
6. Create KPI cards and calculated fields.
7. Develop Tableau Story points to explain project insights.

Non-Functional Requirements

- The dashboard should be easy to use.
- Dashboard loading time should be minimal.
- Data visualizations should be accurate and reliable.
- The system should support future datasets.
- The dashboard should be available for analysis whenever required.

3.3 Data Flow Diagram:

The Data Flow Diagram (DFD) represents the movement of data from the input stage to the final dashboard. It illustrates how raw EV datasets are processed and transformed into meaningful visualizations.

Data Flow

EV Population Dataset

EV Specifications Dataset

EV Charging Stations Dataset

Global EV Sales Dataset



Data Collection



Data Cleaning & Preprocessing

(Python / Excel)



Processed Dataset



Tableau Desktop

(Data Integration & Analysis)



Interactive Dashboard



End User

3.4 Technology Stack:

The Technology Stack describes the software, programming languages, libraries, and tools used to develop the Electric Vehicle Charge and Range Analysis Dashboard.

Component	Technology Used
Programming Language	Python
Data Processing	Pandas, NumPy
Visualization Tool	Tableau Desktop
Data Format	CSV Files
Database	CSV / MySQL (Optional)

Component	Technology Used
Operating System	Windows 10/11
Development Tool	Visual Studio Code
Documentation	Microsoft Word

Python is used for cleaning and preprocessing the datasets. Pandas and NumPy handle data manipulation and analysis. Tableau Desktop is used to build interactive dashboards and story points. The datasets are stored in CSV format and processed locally on a Windows system. Microsoft Word and PowerPoint are used for preparing the project documentation.

4. PROJECT DESIGN:

4.1 Problem–Solution Fit:

Problem:

Users struggle to compare EV performance due to scattered information.

Solution:

Develop an interactive Tableau dashboard integrating four datasets to provide meaningful visual insights.

Benefits:

- Faster analysis
- Better decisions
- Easy comparison
- Interactive visualization

4.2 Proposed Solution:

The proposed solution is a Tableau dashboard that integrates multiple EV datasets into one analytical platform. The dashboard includes KPIs, charts, filters, maps, and story

points, enabling users to compare electric vehicle specifications, charging stations, and sales trends efficiently.

4.3 Solution Architecture:

Flow:

Datasets



Python Data Cleaning



Processed CSV



Tableau Desktop



Dashboard



End Users

5. PROJECT PLANNING & SCHEDULING:

Project Planning and Scheduling is an essential phase that helps organize project activities, allocate responsibilities, estimate effort, and monitor progress throughout the development process. In this project, Agile methodology was followed by dividing the work into multiple sprints. Each sprint focused on completing a specific set of tasks, including data collection, preprocessing, dashboard development, testing, and documentation. This approach ensured systematic development and timely completion of the project.

5.1 Product Backlog:

The Product Backlog is a prioritized list of features and tasks required to successfully complete the project. Each task is represented as a user story and assigned story points based on its complexity and effort.

Sprint	Functional Requirement	User Story	Story Points	Priority
Sprint-1	Data Collection	Collect four EV datasets from reliable sources.	2	High
Sprint-1	Data Cleaning	Remove missing values and duplicate records.	3	High
Sprint-1	Data Preprocessing	Prepare datasets for Tableau visualization.	2	High
Sprint-2	Data Integration	Establish relationships between multiple datasets.	2	High
Sprint-2	Dashboard Development	Design interactive dashboards with KPIs and charts.	5	High
Sprint-3	Dashboard Visualization	Create bar charts, pie charts, maps, scatter plots, tree maps, and line charts.	4	High
Sprint-3	Story Creation	Develop Tableau Story for presenting project insights.	3	Medium
Sprint-4	Testing	Verify dashboard functionality and data accuracy.	2	Medium
Sprint-4	Documentation	Prepare final report and presentation.	3	High
Sprint-4	Final Submission	Complete project review and submit deliverables.	2	High

5.2 Sprint Planning:

The project was divided into four sprints between **30 June 2026 and 14 July 2026**.

Sprint	Duration	Activities
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Sprint-1	30 Jun – 03 Jul	Data collection, preprocessing, and cleaning
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Sprint	Duration	Activities
Sprint-2	04 Jul – 07 Jul	Data integration and dashboard development
Sprint-3	08 Jul – 10 Jul	Dashboard visualizations and Tableau Story creation
Sprint-4	11 Jul – 14 Jul	Testing, documentation, presentation, and final submission

Each sprint had clearly defined objectives. Completing one sprint before starting the next helped the team maintain quality and track progress effectively.

5.3 Project Tracker:

The Project Tracker records the progress of each sprint by comparing planned work with completed work.

Sprint	Total Story Points	Start Date	End Date	Completed Story Points	Status
Sprint-1	20	30 Jun 2026	03 Jul 2026	20	Completed
Sprint-2	20	04 Jul 2026	07 Jul 2026	20	Completed
Sprint-3	20	08 Jul 2026	10 Jul 2026	20	Completed
Sprint-4	20	11 Jul 2026	14 Jul 2026	20	Completed

The Project Tracker shows that all planned story points were completed within the scheduled duration, indicating successful execution of the project plan.

5.4 Velocity:

Velocity measures the amount of work completed during each sprint.

- **Total Story Points:** 80
- **Total Number of Sprints:** 4
- **Story Points Completed:** 80
- **Average Velocity:** 20 Story Points per Sprint

Velocity Table

Sprint	Story Points Completed
Sprint-1	20

Sprint	Story Points Completed
Sprint-2	20
Sprint-3	20
Sprint-4	20

Average Velocity 20

Explanation

The team's average velocity was **20 story points per sprint**, indicating consistent progress throughout the project. Maintaining a stable velocity helped complete all planned tasks within the project timeline.

5.5 Burndown Chart:

Date	Remaining Story Points
30 Jun	80
01 Jul	74
02 Jul	68
03 Jul	60
04 Jul	54
05 Jul	48
06 Jul	40
07 Jul	34
08 Jul	28
09 Jul	25
10 Jul	20
11 Jul	15
12 Jul	10

Date Remaining Story Points

13 Jul 5

14 Jul 0

6. FUNCTIONAL AND PERFORMANCE TESTING:

Functional and Performance Testing is carried out to verify that the Electric Vehicle Charge and Range Analysis Dashboard performs as expected. Functional testing ensures that all dashboard features such as data import, filtering, calculations, visualizations, and story points work correctly. Performance testing evaluates the dashboard's responsiveness, accuracy, and ability to process multiple datasets efficiently.

6.1 Performance Testing:

6.1.1 Data Rendered:

The dashboard uses four electric vehicle datasets that were successfully imported into Tableau Desktop. The datasets include EV Population, EV Specifications, EV Charging Stations, and Global EV Sales. After importing, the data was verified to ensure that all records and attributes were correctly loaded for analysis. The rendered data forms the foundation for creating interactive visualizations and dashboards.

Tableau Public - Book1

File Data Window Help

Connections Add

Cheapestelect...rs-EVDatabase
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

- Cheapestelectr...EVDatabase.csv
- electric_vehicle_station_list.csv
- ElectricCarData_Clean.csv
- EVIndia.csv

[New Union](#)

[New Table Extension](#)

Cheapestelectriccars-EVDatabase

Filters 0 | Add

Cheapestelectriccars-EVD... 46 fields 497 rows 100 rows

Name Cheapestelectriccars-EVDatabase.csv

Description No description available.

Fields

Type	Field Name	Physical Table	Rem...
Alt	Name	Cheapestelectric...	Name
Alt	Subtitle	Cheapestelectric...	Subtitle

Alt	Alt	Alt	Alt
Cheapestelectriccars-EVDatabase.csv	Cheapestelectriccars-EVDatabase.csv	Cheapestelectriccars-EVDatabase.csv	Cheapestelectriccars-EVDatabase.csv
Name	Subtitle	Acceleration	Top Speed
Opel Ampera-e	Battery Electric Vehicle ...	7.3 sec	150 km/h
Renault Kangoo Maxi ZE 33	Battery Electric Vehicle ...	22.4 sec	130 km/h
Nissan Leaf	Battery Electric Vehicle ...	7.9 sec	144 km/h
Audi e-tron Sportback 55 qu...	Battery Electric Vehicle ...	5.7 sec	200 km/h
Porsche Taycan Turbo S	Battery Electric Vehicle ...	2.8 sec	260 km/h
Nissan e-NV200 Evalia	Battery Electric Vehicle ...	14.0 sec	123 km/h

Data Source EV brands Brands according to body style Brand filtered by PowerTrain type Dashboard 1 Story of Electric cars in India No. of Models each by brand Different brands of EV cars glob... Different brands of EV cars in In...

6.1.2 Data Preprocessing:

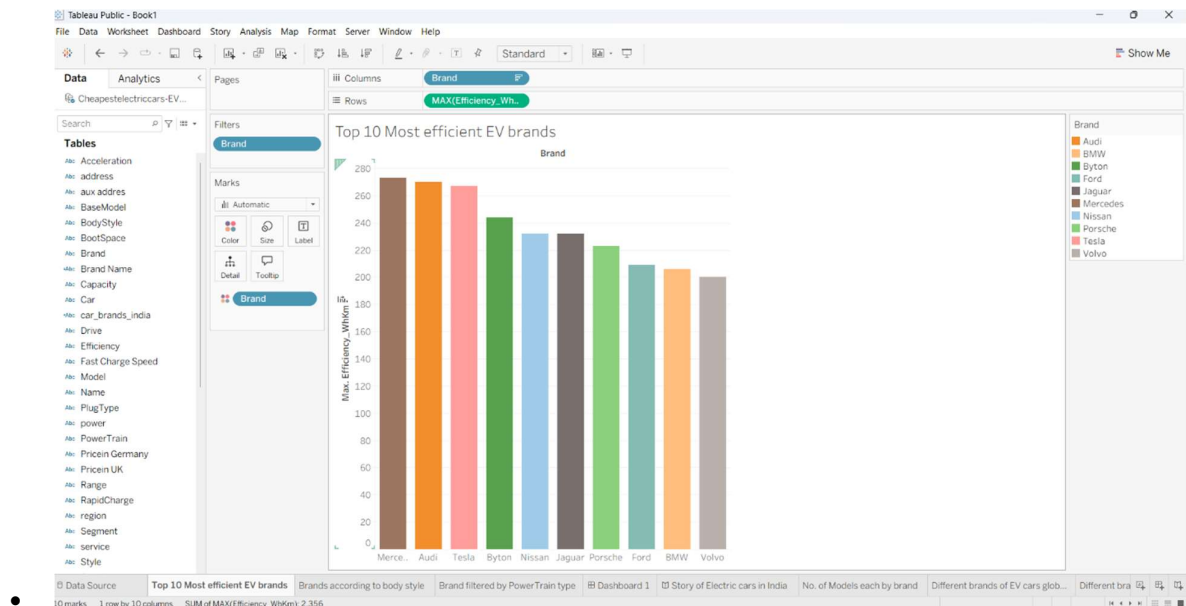
Before visualization, the datasets were cleaned and preprocessed to improve data quality. Missing values and duplicate records were removed, data types were corrected, and unnecessary columns were eliminated. Relationships between the datasets were established to enable integrated analysis. These preprocessing steps ensured accurate and reliable dashboard results.

6.1.3 Utilization of Filters:

Interactive filters were implemented to allow users to analyze data based on different criteria. Filters improve dashboard usability by enabling quick comparisons and customized analysis.

Filters Used

- Brand
- Vehicle Model
- Vehicle Type
- Model Year
- Battery Capacity
- Electric Range
- Charging Time
- Country/Region (if available)



6.1.4 Calculated Fields Used:

Calculated fields were created to generate useful business metrics from the datasets. These calculations provide summarized information that helps users understand EV performance and market trends more effectively.

Calculated Fields

- Average Electric Range
- Average Battery Capacity
- Average Charging Time
- Total EV Models
- Total Manufacturers
- Total Charging Stations
- EV Sales Growth Percentage

Tableau Public - Book1

File Data Worksheet Dashboard Story Analysis Map Form

Data | Analytics <

Cheapestelectriccars-EV...

Search

Tables

- Acceleration
- address
- aux addres
- BaseModel
- BodyStyle
- BootSpace
- Brand
- Brand Name
- Capacity
- Car
- car_brands_india
- Drive
- Efficiency
- Fast Charge Speed
- Model
- Name
- PlugType
- power
- PowerTrain
- Pricein Germany
- Pricein UK
- Range
- RapidCharge
- region
- Segment
- service
- Style

Pages

Filters

BodyStyle

Marks

Circle

Color Size Label

Detail Tooltip

AGG(Bodystyle..)

BodyStyle

Brand

BodyStyle

Data Source | Top 10 Most efficient EV brands | Brands

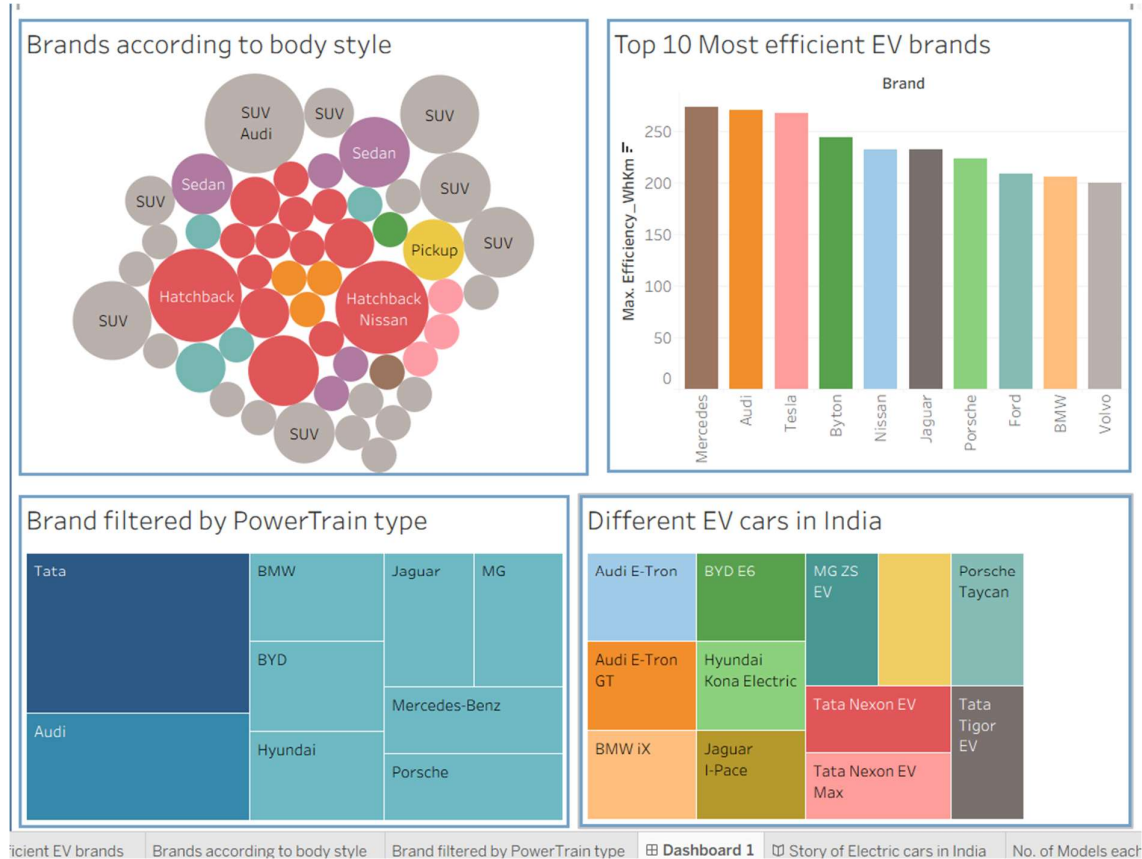
52 marks 1 row by 1 column SUM of AGG(Bodystyle_count): 103

6.1.5 Dashboard Design:

The dashboard was designed using multiple visualizations to provide clear and interactive insights into electric vehicle data. KPI cards summarize important information, while charts and maps enable detailed analysis.

Visualizations Used

- KPI Cards
- Bar Chart – EV Models by Brand
- Pie Chart – Vehicle Type Distribution
- Line Chart – Global EV Sales Trend
- Scatter Plot – Battery Capacity vs. Electric Range
- Tree Map – Brand-wise EV Models
- Heat Map – Charging Time Analysis
- Map – Charging Station Distribution
- n

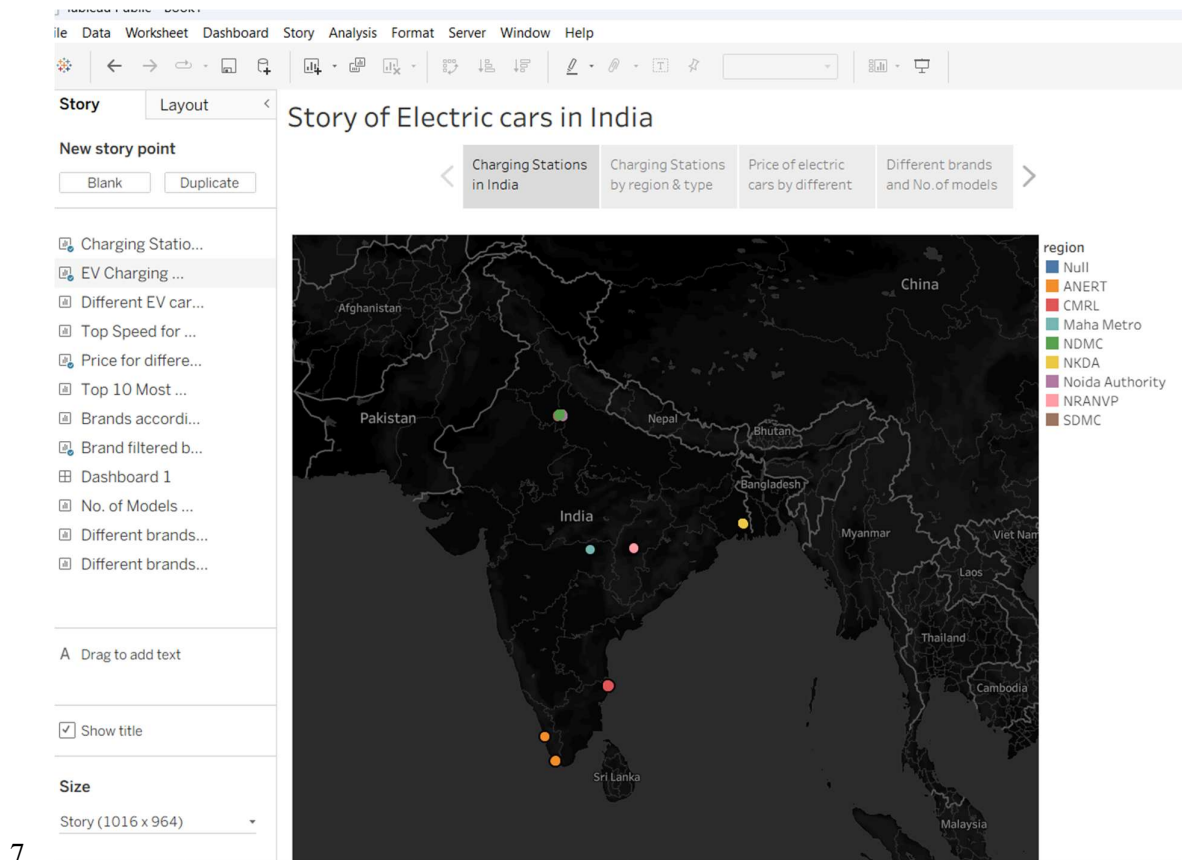


6.1.6 Story Design:

Tableau Story was created to present the project findings in a logical sequence. Each story point focuses on a specific aspect of the analysis, making it easier for users to understand the dashboard insights.

Story Points

1. Project Overview
2. EV Population Analysis
3. Brand and Model Comparison
4. Battery Capacity and Electric Range Analysis
5. Charging Station Analysis
6. Global EV Sales and Final Insights



Performance Summary:

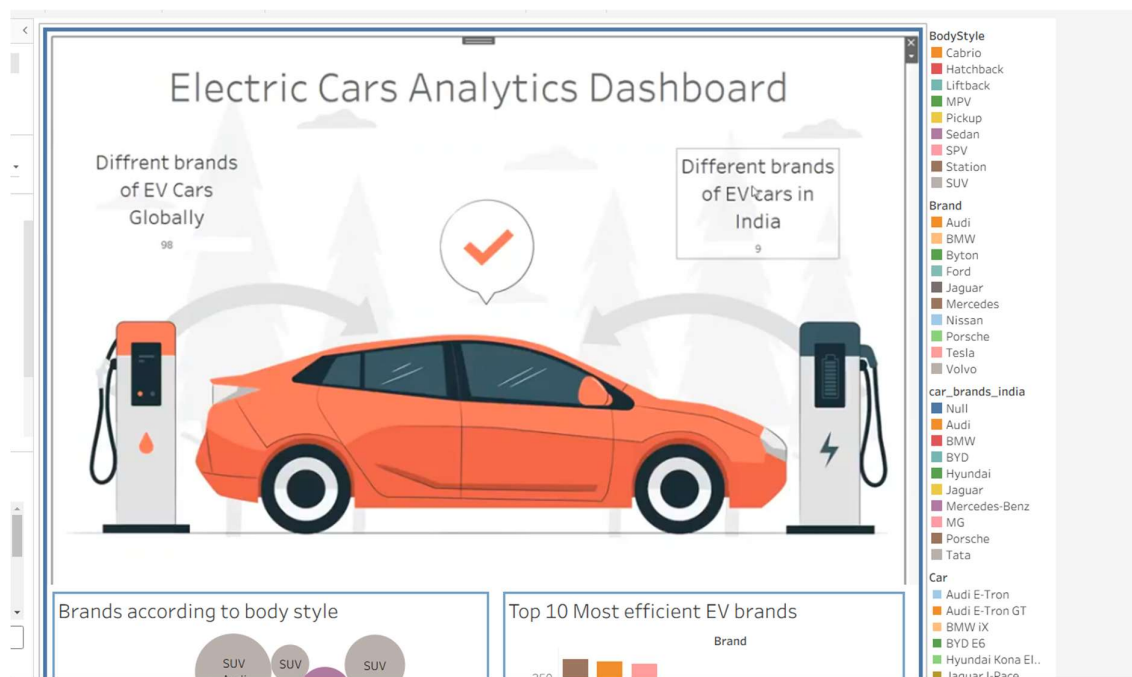
- The dashboard successfully processed all four datasets and generated interactive visualizations without significant delays. Filters responded efficiently, calculated fields produced accurate results, and the dashboard provided

smooth navigation between visualizations and story points. The testing confirmed that the system met both functional and performance requirements and delivered reliable insights for electric vehicle analysis.

7.RESULTS:

The results of the project demonstrate the successful implementation of the Electric Vehicle Charge and Range Analysis Dashboard using Tableau Desktop. The dashboard integrates four electric vehicle datasets and presents meaningful insights through interactive charts, maps, KPI cards, filters, and story points. The visualizations help users analyze electric vehicle performance, charging infrastructure, battery specifications, and market trends efficiently.

Figure 7.1: Dashboard Home Page:



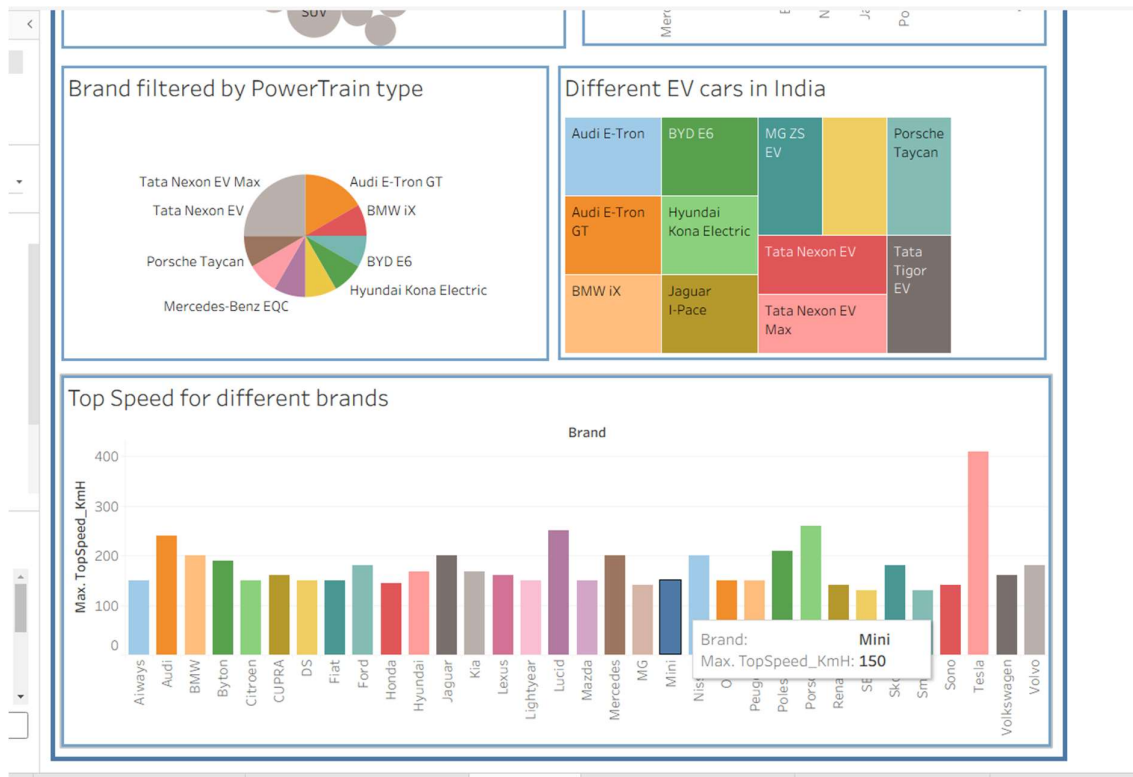


Figure 7.2: Charging Stations By region in India:



Figure 7.3: EV Charging Stations Map in India:

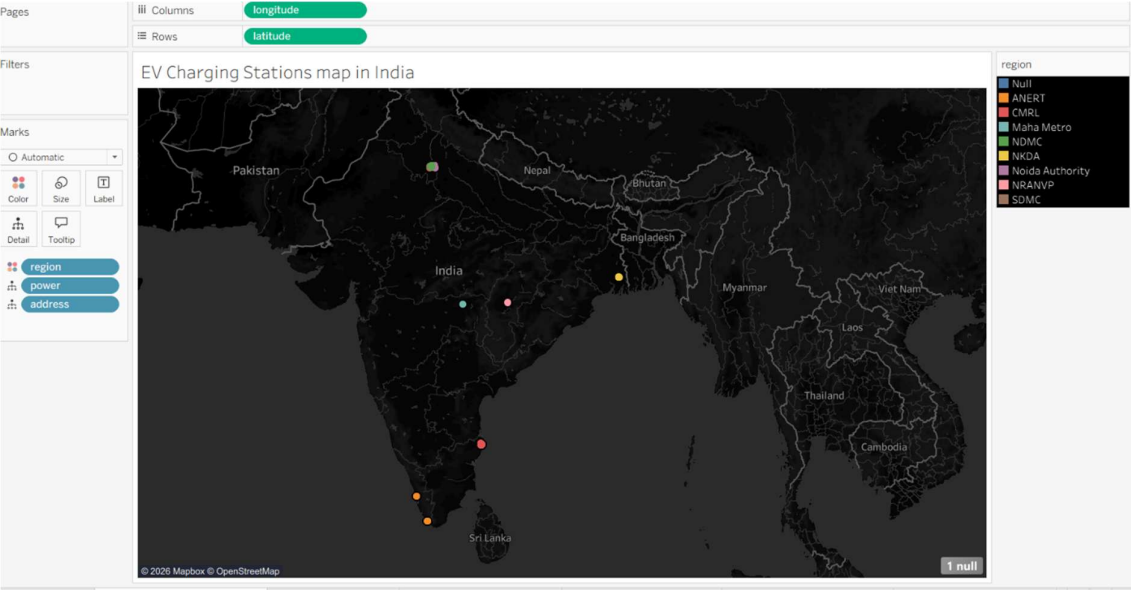


Figure 7.4: Different EV Cars in India:



Figure 7.5: Top Speed for Different brands:

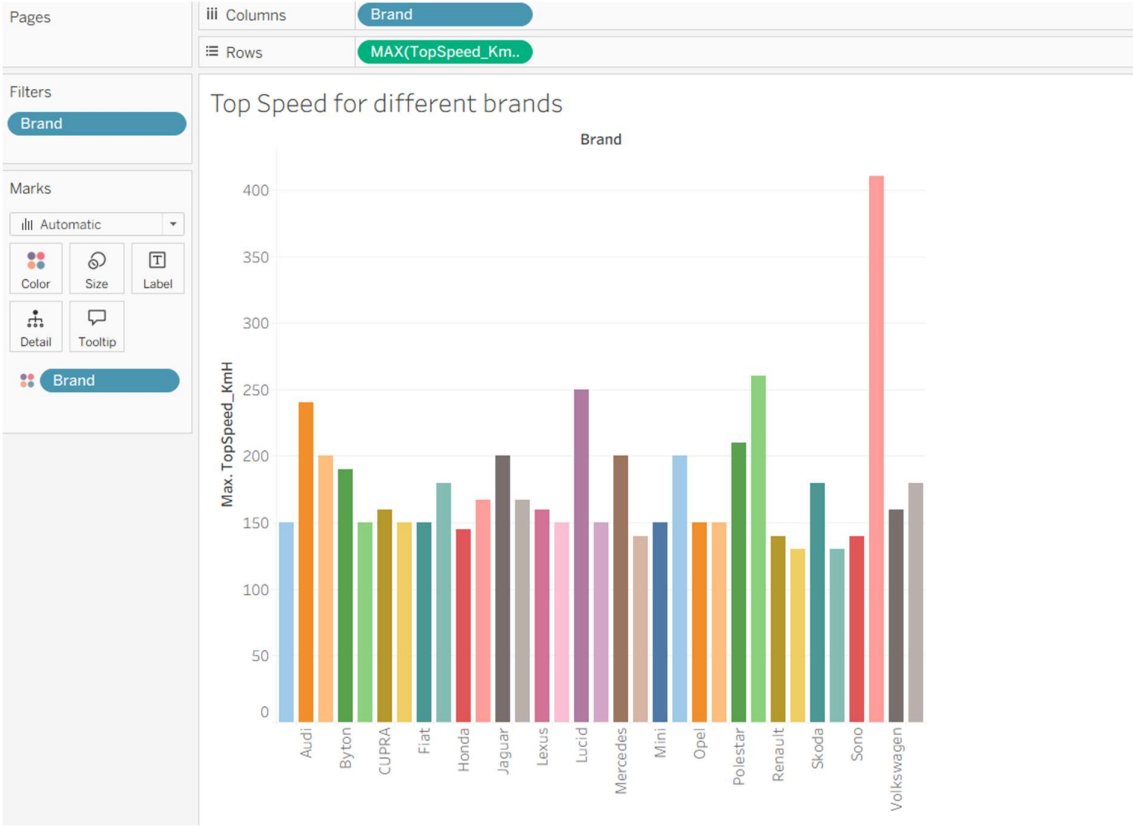


Figure 7.6: Brands According to Body Style:

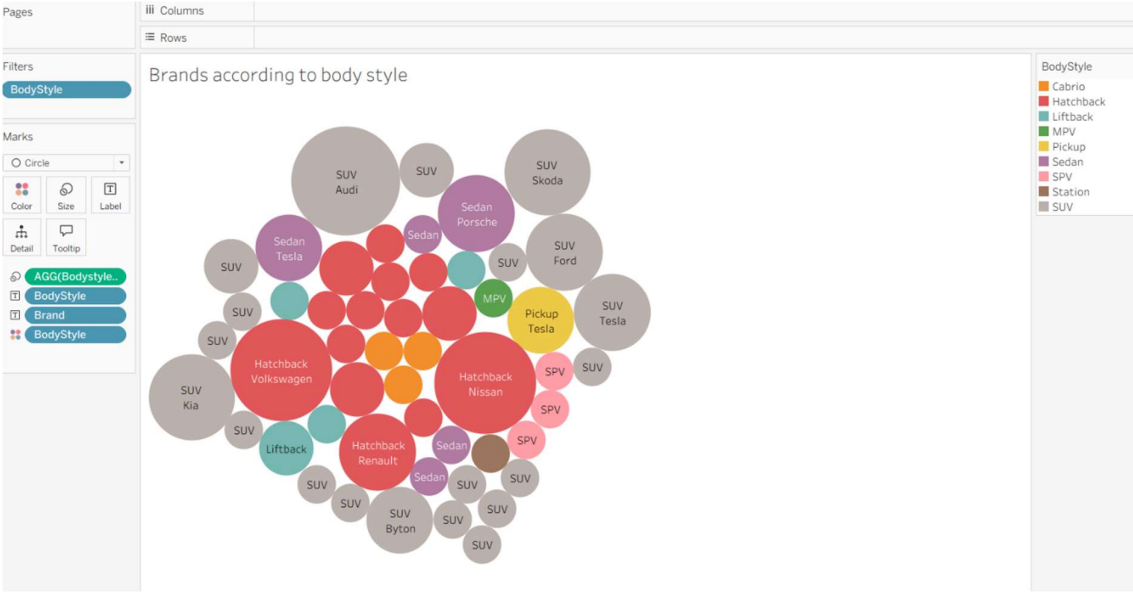


Figure 7.7: No. of Models each by brand:

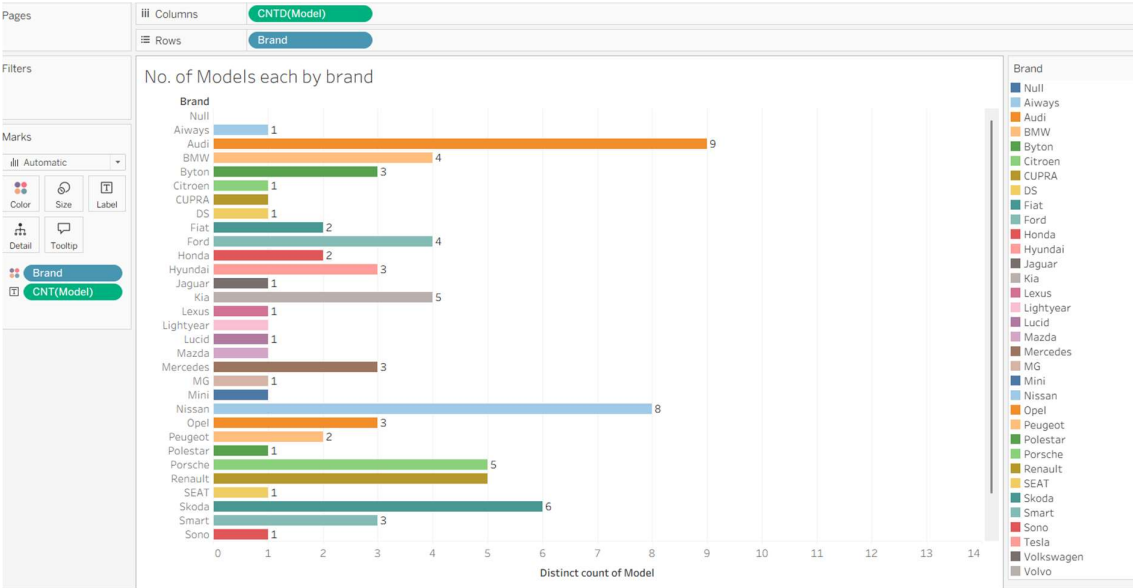
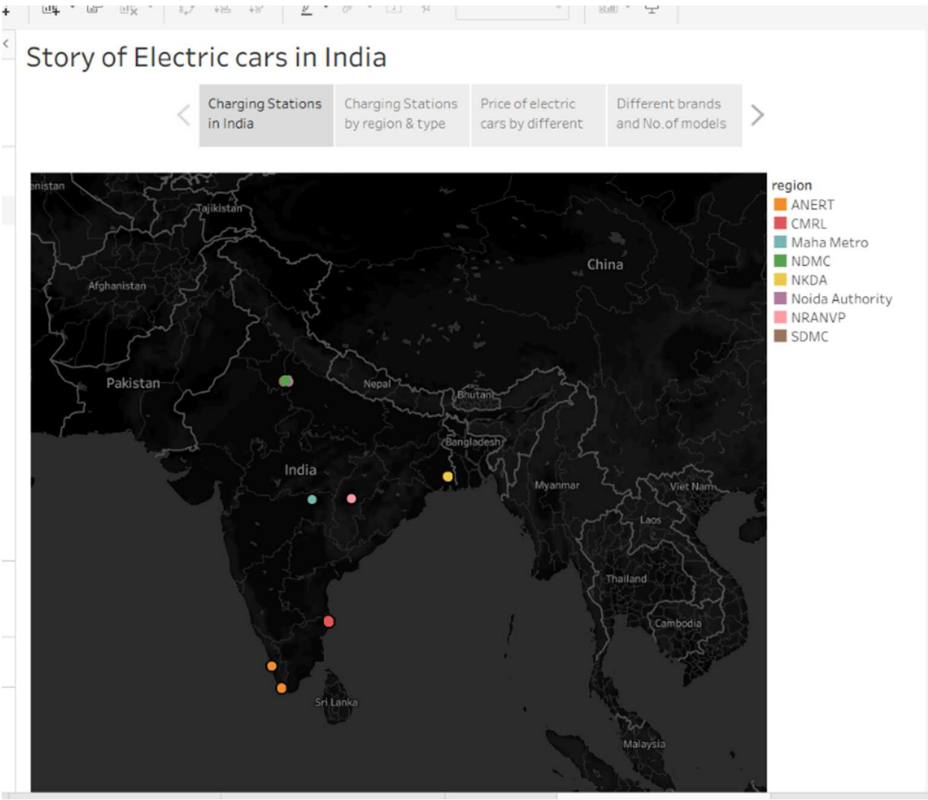
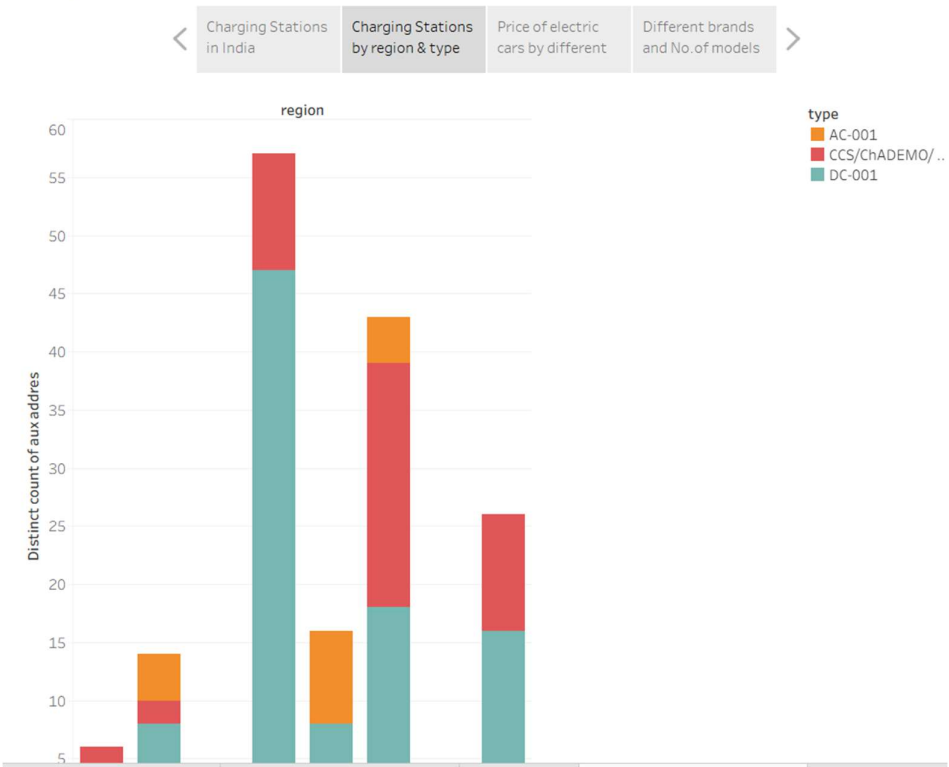


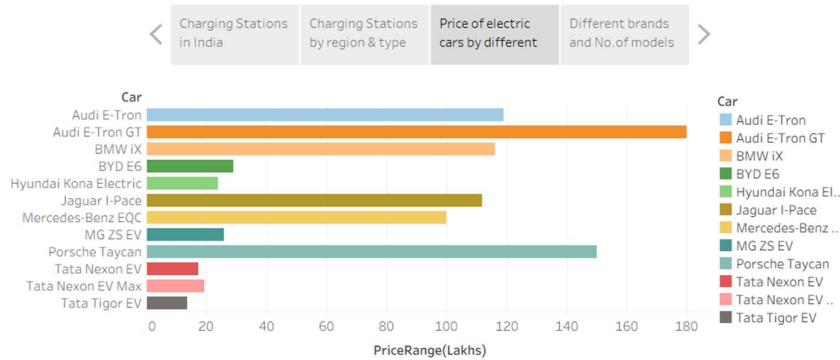
Figure 7.8: Story of Electric Cars in India:



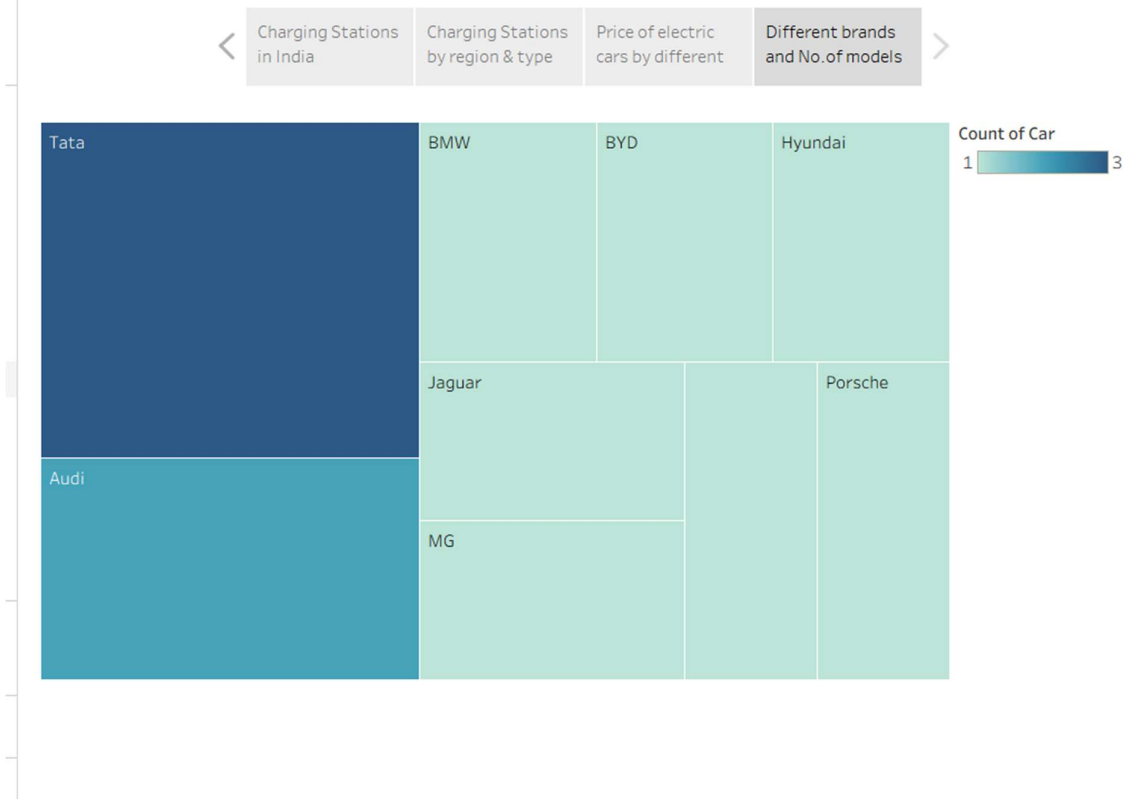
Story of Electric cars in India



Story of Electric cars in India



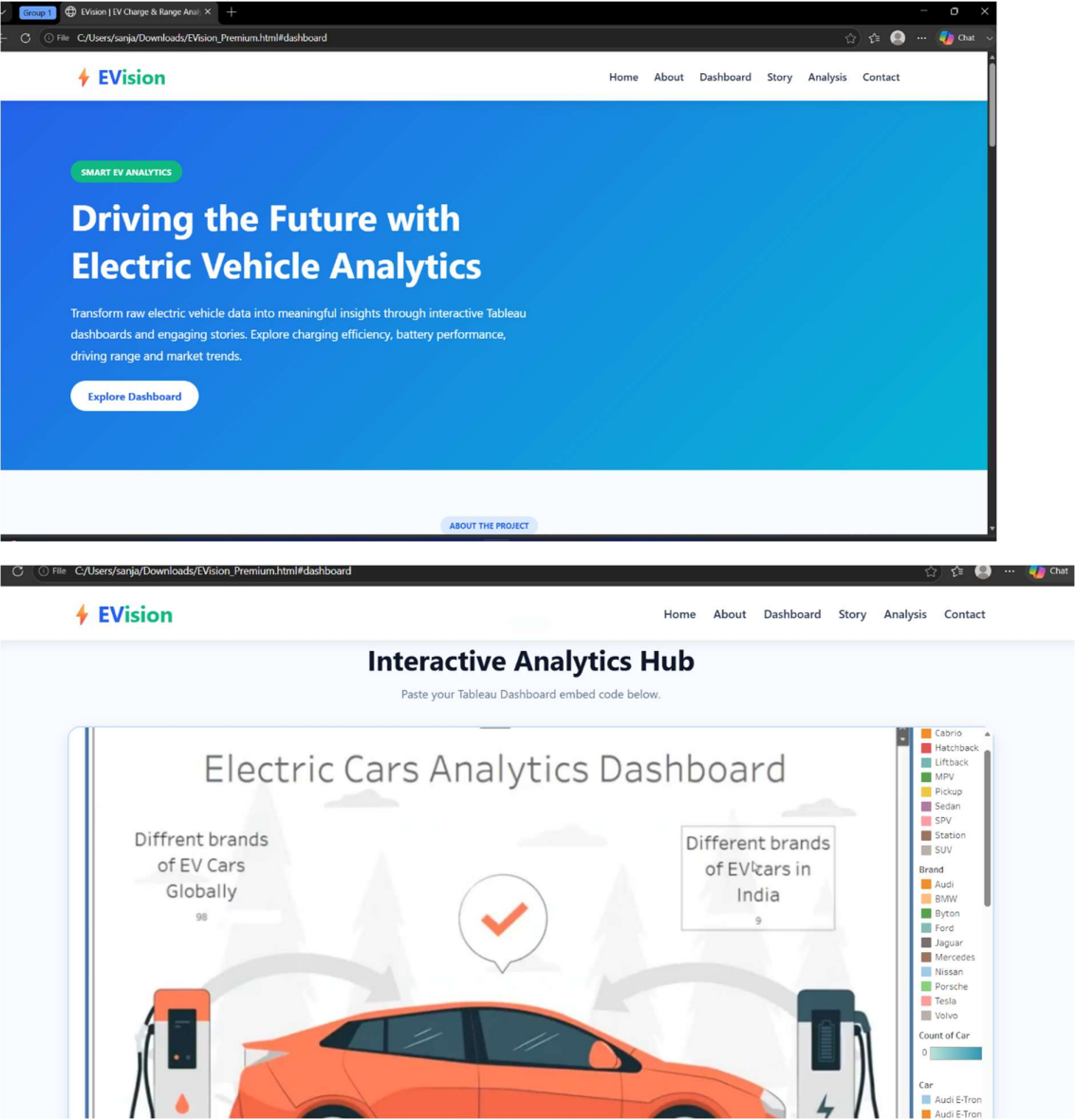
Story of Electric cars in India

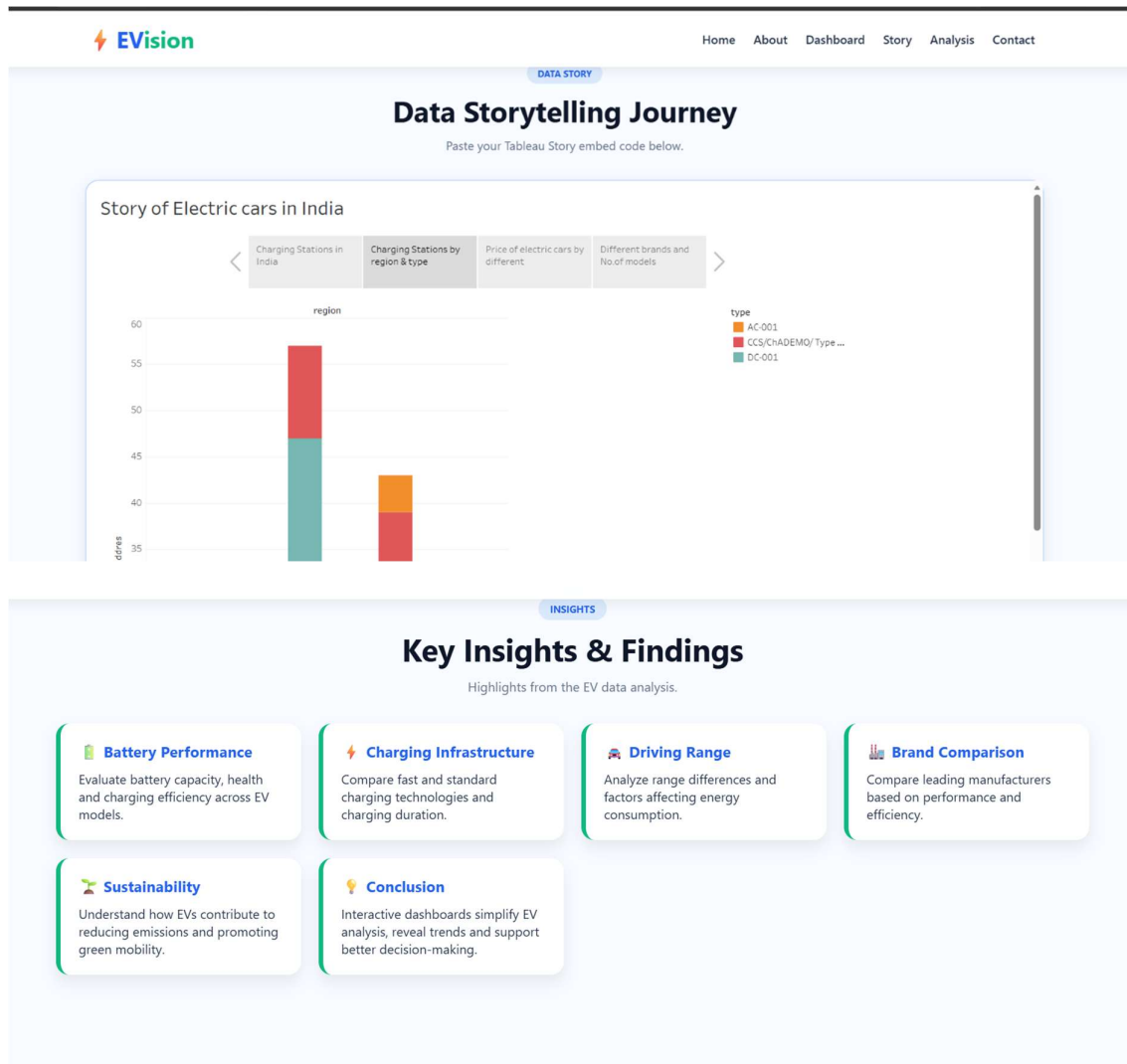


The developed dashboard successfully integrates four electric vehicle datasets into a single interactive platform. The visualizations, KPI cards, filters, and story points enable users to compare EV brands, analyse battery capacity, evaluate charging infrastructure, and study market trends. The results demonstrate that Tableau provides an effective

solution for transforming raw electric vehicle data into meaningful insights, supporting informed decision-making for EV buyers, researchers, and industry stakeholders.

Figure 7.9: Web Integration:





8. ADVANTAGES & DISADVANTAGES

8.1 Advantages:

1. Interactive Data Visualization

The dashboard presents complex electric vehicle data through interactive charts, maps, KPI cards, and filters, making the information easy to understand.

2. Centralized Data Analysis

It combines four different EV datasets into a single platform, allowing users to access and analyze all relevant information without switching between multiple sources.

3. Easy Comparison of Electric Vehicles

Users can compare EV brands, battery capacity, electric range, charging time, and sales trends using interactive visualizations.

4. Improved Decision-Making

The dashboard provides meaningful insights that help customers, researchers, and manufacturers make informed decisions regarding electric vehicle selection and market analysis.

5. User-Friendly Interface

The Tableau dashboard is simple to navigate and allows users to explore data using filters and interactive controls without requiring technical expertise.

6. Time-Saving

Instead of manually analyzing spreadsheets, users can quickly obtain insights through graphical representations and KPI summaries.

7. Better Data Interpretation

Visualizations such as bar charts, line charts, scatter plots, tree maps, and heat maps make it easier to identify trends, patterns, and relationships within the data.

8. Scalable Solution

The dashboard can be enhanced by adding new datasets, additional visualizations, and advanced analytics without changing the existing structure.

8.2 Disadvantages:

1. Dependency on Dataset Quality

The accuracy of the dashboard depends entirely on the quality and completeness of the input datasets. Incorrect or missing data can affect the analysis.

2. No Real-Time Data Integration

The current dashboard uses static datasets and does not provide real-time updates for electric vehicle information or charging station availability.

3. Limited Predictive Analysis

The project focuses on descriptive analysis and visualization rather than forecasting future trends using machine learning models.

4. Software Dependency

Users need Tableau Desktop or Tableau Public to access and interact with the dashboard, which may limit accessibility in some environments.

5. Performance with Large Datasets

As the volume of data increases, dashboard loading time and responsiveness may decrease if the system resources are limited.

6. Limited Geographic Coverage

The dashboard analysis is restricted to the information available in the selected datasets and may not represent all countries or regions.

7. Manual Dataset Updates

Whenever new EV data becomes available, the datasets must be manually updated and refreshed to keep the dashboard current.

9. CONCLUSION

The **Electric Vehicle Charge and Range Analysis Dashboard** was successfully developed using Tableau Desktop to provide an interactive platform for analyzing electric vehicle data. The project integrates four different datasets related to EV population, vehicle specifications, charging stations, and global EV sales into a single dashboard. Through data preprocessing, visualization, and interactive filtering, the dashboard enables users to compare electric vehicle brands, battery capacity, driving range, charging time, and market trends efficiently.

The dashboard simplifies complex datasets into meaningful visualizations using KPI cards, bar charts, line charts, scatter plots, pie charts, tree maps, and maps. Tableau Story further enhances the user experience by presenting the analysis in a structured sequence. The project demonstrates how Business Intelligence tools can transform raw data into actionable insights that support informed decision-making.

Overall, the project achieved its objectives by developing a user-friendly and interactive dashboard that helps customers, researchers, manufacturers, and policymakers understand electric vehicle performance and market trends. The solution highlights the importance of data visualization in supporting sustainable transportation and promoting the adoption of electric vehicles.

10. FUTURE SCOPE

The current project provides descriptive analysis of electric vehicle data using Tableau. In the future, several enhancements can be implemented to improve its functionality and usefulness.

- Integrate real-time electric vehicle and charging station data using APIs.
- Develop a web-based dashboard that can be accessed from any device.
- Create a mobile application for convenient access to EV analytics.
- Implement machine learning models to predict EV sales and future market trends.
- Develop an EV recommendation system based on user preferences and driving requirements.
- Include battery health analysis and charging efficiency prediction.
- Integrate live traffic and route planning with nearby charging stations.
- Add carbon emission savings analysis to measure environmental benefits.
- Expand the dashboard by including additional countries, manufacturers, and new electric vehicle models.
- Publish the dashboard on Tableau Public or Tableau Server for wider accessibility.

These enhancements will make the dashboard more intelligent, scalable, and beneficial for both individual users and organizations.

11. APPENDIX

11.1 Source Code:

The project uses Tableau Desktop for dashboard development. Data preprocessing was performed using Python and Microsoft Excel before importing the datasets into Tableau.

Example Python Libraries Used:

- Python
- Pandas
- NumPy
- Tableau Desktop

11.2 GitHub Repository:

11.3 Project Demo Video: